PROPELLUS · COMPLIANCE INTELLIGENCE

Alpha Surveillance

AI-powered detection, alerting and compliance for operations that cannot afford to miss anything.

PREPARED FOR

[Client Name]

Presented by Propellus · October 2026

Commercial in confidence

THE STAKES

The cost of what nobody was watching.

Recording is not monitoring. These are the losses that accumulate while footage sits unwatched.

$181.4B

Annual cost of work injuries

Wage and productivity losses, medical and administrative costs borne by US employers in a single year.

5,070

Fatal work injuries in 2024

1,032 of them in construction and extraction occupations alone — a rate of 3.3 per 100,000 workers.

$48,000

Cost per medically consulted injury

Before counting citations, downtime, insurance loading or the cost of defending a claim.

102M

Working days lost

Productive capacity erased in one year by incidents that were, in most cases, preventable.



Every one of these figures began as a moment on camera that nobody happened to be watching.

Sources: National Safety Council, Injury Facts (2024 data); U.S. Bureau of Labor Statistics, Census of Fatal Occupational Injuries (2024).

THE HUMAN LIMIT

Experience doesn't fix it. Attention simply runs out.

66%

of trained operators missed an event in plain sight.

In a controlled study of 171 CCTV operators watching optimal-quality footage on a single screen, two thirds failed to notice a salient, unexpected event. The 84 participants with up to 19 years of professional monitoring experience performed no better than the 87 with none.

Missed-detection rate by event type



"Prior task experience did not inoculate operators against inattentional blindness."

Source: PLOS ONE, "Pirate Stealth or Inattentional Blindness? The Effects of Target Relevance and Sustained Attention on Security Monitoring for Experienced and Naïve Operators." n = 171.

THE GAP

Four failures every recording-only site shares

Detection happens after the fact

Footage becomes evidence only once an incident forces someone to go looking for it. By then the injury, the loss or the citation is already on the books.

Coverage scales, attention doesn't

Every camera added widens the gap between what is recorded and what is genuinely seen. Staffing to close that gap is the one cost that rises with every install.

Compliance rests on paperwork

Checklists capture what someone remembered to write down, not what happened on the floor. Auditors and insurers increasingly decline to treat the two as equivalent.

Exposure without evidence

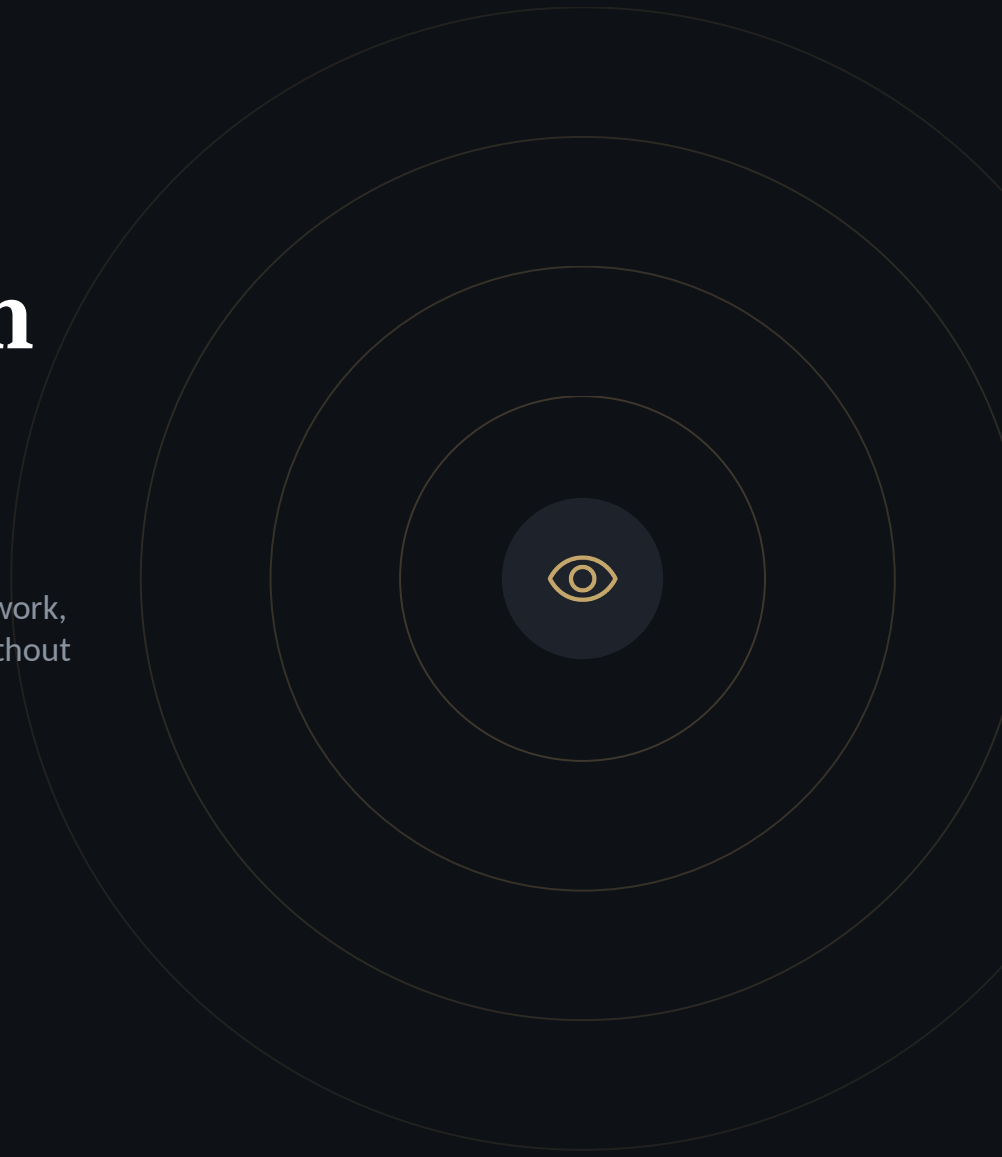
OSHA penalties reach $16,550 per serious violation and $165,514 per willful or repeated one. Contesting a citation without timestamped proof begins from behind.

Penalty maximums effective for citations issued after 15 January 2026. Source: U.S. Occupational Safety and Health Administration.

THE PLATFORM

Alpha Surveillance turns the cameras you already own into an always-on compliance officer.

It connects to existing RTSP and IP cameras, runs AI inference on an edge device inside your own network, and evaluates every frame against the Standard Operating Procedures you define — continuously, without fatigue, and without depending on a person looking at the right screen at the right moment.



- Flags the violation as it happens — not on review, not next quarter
- Produces a timestamped, evidence-backed record of every single event
- Authored around your SOPs, not a fixed vendor-defined rule set

WHAT IT DOES

Three pillars, one platform



Detection

Vision models and connected sensors evaluate conditions continuously — missing PPE, restricted-zone entry, unsafe proximity to equipment, or any rule you can describe in plain language.



Alerting

The moment a rule breaks, the event reaches the people who can act on it — pushed live to the operator console and routed by email, with per-camera cooldowns so one recurring condition never floods the channel.



Compliance

Every detection is written to an append-only journal with its evidence frame, timestamp, source and originating rule — producing the defensible record that audits, insurers and incident reviews actually require.

THE PIPELINE

From camera feed to defensible record

Five stages, each independently scalable. The loop closes in seconds, not on the next review cycle.

01



Ingest

Existing RTSP or IP cameras stream to the on-site edge device. Video never leaves your network to be analysed.

02



Infer

Frames are evaluated locally against every SOP rule mapped to that specific camera and zone.

03



Alert

A confirmed violation pushes instantly to the console and to the recipients who own that site.

04



Record

Evidence frame, timestamp, camera and rule are committed to the append-only audit journal.

05



Analyse

Events roll up into trends, spatial hotspots and per-site compliance scoring over time.



One camera or one thousand. Because the stages scale independently, adding cameras adds capacity rather than queue time.

Detection-to-alert latency depends on model, stream resolution and edge hardware, and is confirmed during the site survey.

THE DETECTION LAYER

Three model strategies, one rule engine

Because no single model covers everything a working site actually needs to enforce.

Purpose-trained models

Detectors trained on footage from your own site for the conditions that matter most to you — used where generic, off-the-shelf accuracy is not good enough to act on.

Specialist hosted models

Production-grade detectors for well-defined domains such as construction PPE, covering hard hat, hi-vis vest and mask compliance from day one.

Open-vocabulary grounding

Describe a new target in plain language and the model locates it without a retraining cycle — turning new rules into a configuration change, not a project.

Why this matters

You are never locked to a fixed label list. As your SOPs evolve, the rule engine evolves with them — cameras already installed pick up new detections through configuration rather than reinstallation.

Mapped to the standards regulators actually cite

Six of the ten most frequently cited OSHA standards in FY 2025 describe conditions a camera can verify.

STANDARD	OSHA TOP-10 ENTRY	WHAT ALPHA SURVEILLANCE DETECTS
1926.501	Fall protection, general requirements **#1**	Presence inside unprotected edge zones, missing harness anchorage, entry to fall-risk areas
1910.134	Respiratory protection **#5**	Respirator or mask absent within a designated respiratory-protection zone
1926.451	Scaffolding, construction **#6**	Unsafe scaffold access, personnel on incomplete platforms, overloading indicators
1926.503	Fall protection training **#7**	Unqualified personnel in fall-risk zones, cross-checked against enrolment records
1910.178	Powered industrial trucks **#8**	Pedestrian–forklift proximity, unsafe travel paths, obstructed aisles
1926.102	Eye and face protection **#9**	Eye or face protection absent at designated task areas

OSHA publishes its Top 10 by rank only; citation counts are not released alongside the list. Source: OSHA, Top 10 Most Frequently Cited Standards, FY 2025 (1 Oct 2024 – 30 Sep 2025).

MACHINE & ENVIRONMENTAL TELEMETRY

Vision is one sensor. It isn't the only one.

Industrial sensors stream into the same edge device, the same alert routing and the same audit record as every camera on site.

Water pressure & flow

Line pressure, flow rate and pump load. A sustained drop signals a leak or blockage; a spike signals regulator or valve failure before a line ruptures.

< 2.1 bar, or Δ15% within 60 s

Air purity

Particulate, gas and VOC concentration measured against occupational exposure limits, so a breach raises an alert while the space can still be cleared.

PM2.5 > 35 µg/m³ sustained

Fire & thermal rise

Smoke, flame signature and abnormal thermal rise on machinery — caught at the sensor and cross-confirmed on the nearest camera before anyone is dispatched.

ΔT > 20 °C above baseline

Electrical fluctuation

Voltage sag, surge, phase imbalance and current draw: the earliest signature of motor degradation, and of the unplanned downtime that follows it.

Sag > 10% held beyond 2 s

One pipeline, whatever the sensor. A pressure breach and a missing hard hat follow an identical path — same routing, same immutable record, same compliance view.

Threshold values shown are configurable examples, not product defaults. PM2.5 reference: U.S. EPA 24-hour PM2.5 NAAQS, 35 µg/m³ (retained in the 2024 final rule).

ALERTING

An alert that reaches nobody is not an alert

Detection is the easy half. Delivery under real-world conditions is where systems quietly fail.

Live operator push

Violations appear on the console in real time over a persistent connection — no refresh, no polling, no waiting on a scheduled report.

Cooldown control

Per-source, per-rule suppression windows stop a single persistent condition from generating hundreds of duplicate alerts and training staff to ignore them.

Routed notification

Email delivery to the roles that own each site, so an alert still reaches a supervisor when nobody is sitting at the console.

Guaranteed handoff

Alerts are queued and retried through a durable outbox, so a transient network or service fault delays delivery rather than losing the event.

Alert fatigue is the failure mode that kills these systems. Cooldowns, routing rules and delivery guarantees are what keep an alert meaningful six months after go-live — not just on day one.

COMPLIANCE

Evidence that survives an audit

A violation you cannot prove is a liability, not a control.

Immutable journaling

Events are appended, never edited or deleted. The record of what was detected cannot be quietly revised after the fact.

Evidence attached at capture

The frame or sensor reading that triggered the violation is stored with the event, so the proof and the claim can never drift apart.

Tenant isolation

Every record is scoped to your organisation and enforced at the query layer — not merely hidden in the interface.

Queryable retention

Purpose-built time-series storage keeps long audit histories searchable instead of archived into inaccessibility.

WHAT AN AUDITOR RECEIVES

- The exact frame or reading, at the exact second
- Camera or sensor, zone, location and site
- The SOP and rule version in force at the time
- Detection confidence and model identifier
- Who was notified, and when it was acknowledged
- A complete, exportable event history

ADD-ON MODULES

Extend the platform as the operation grows

Every module runs on the cameras and the edge device already installed — no second rollout.

Attendance

Vision-based check-in, check-out and live headcount. No badges, turnstiles or biometric hardware to buy and maintain.

Analytics & heatmaps

Violation trends, spatial hotspots and risk scoring by site, camera, shift or individual SOP.

Human re-identification

Follow the same individual across multiple camera views to track movement and cumulative repeat violations.

Sensor telemetry

Water, air, thermal and electrical monitoring brought into the same alerting and compliance record.

Visitor analytics

Footfall, dwell, queue and demographic insight from the cameras already enforcing your rules.

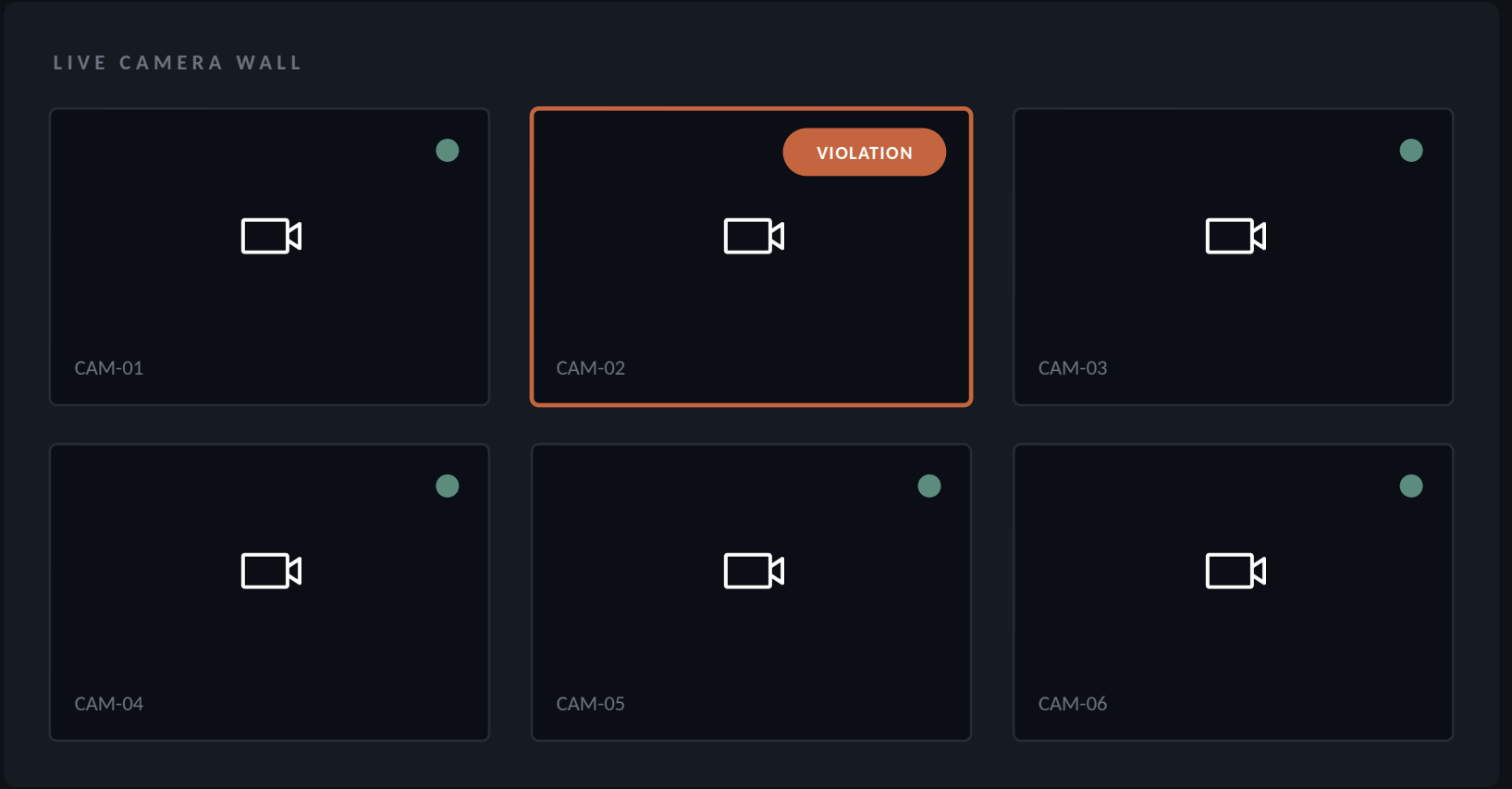
SOP authoring

Define, version and assign detection rules per camera from the admin console, without a deployment.

THE OPERATOR VIEW

One console for every camera, every site

Illustrative concept.



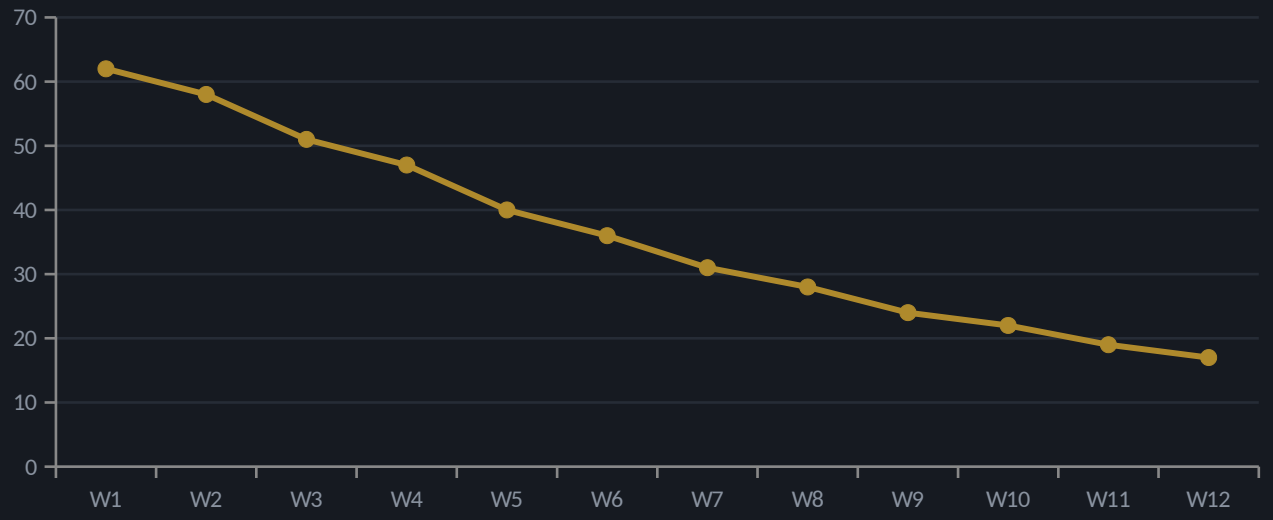
COMPLIANCE ANALYTICS

Compliance becomes a number you can manage

Trend, concentration and repeat-offence patterns — the view a safety or operations lead reports upward.

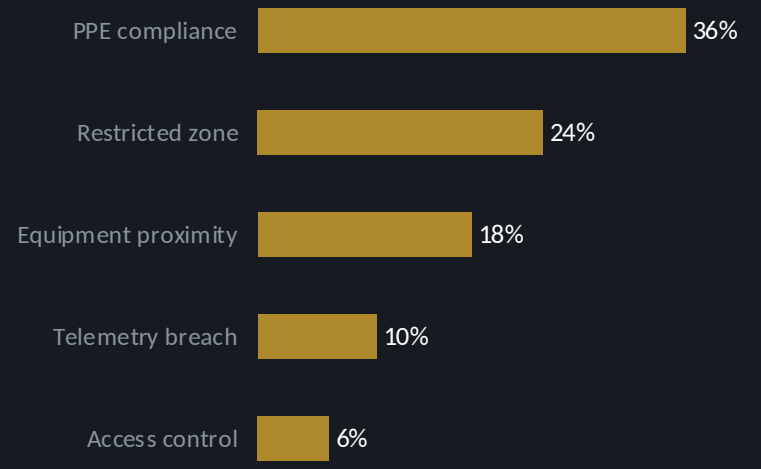
Violations detected per week

Illustrative — a typical enforcement curve after go-live



Where violations concentrate

Illustrative — share of events by category



Figures on this slide are illustrative and provided to demonstrate reporting format. They are not measured client results.

VISITOR & CUSTOMER ANALYTICS

The same cameras also explain your customers

Compliance is the reason the system goes in. Commercial insight is what makes the same hardware pay for itself twice.

Footfall & occupancy

Entries, exits and live occupancy by zone and hour — counted, not inferred from till receipts.

Dwell & engagement

How long people spend in each zone, and which areas or displays genuinely hold attention.

Queue & wait time

Queue length and wait duration at every service point, with alerts when thresholds are breached.

Peak-hour patterns

Demand curves by hour and weekday, so staffing follows real traffic rather than habit.

Age & gender bands

Anonymous, aggregate demographic distribution of visitors — counts and ratios, never identity.

Journey & flow

Zone-to-zone movement paths showing how people actually move through the space.

Aggregate, not identity

Demographic inference runs on the edge device and produces counts and distributions only — no face templates stored, no individual profiles built, retention configurable. In regulated jurisdictions the demographic layer can be switched off entirely while footfall, dwell and queue analytics keep running.

THE DIFFERENCE

Beyond traditional CCTV

TRADITIONAL CCTV	ALPHA SURVEILLANCE
✕ Records for later	✓ **Detects as it happens**
✕ Found on manual review, if at all	✓ **Flagged the instant the rule breaks**
✕ An unstructured footage archive	✓ **An immutable evidence record per event**
✕ Cameras only	✓ **Cameras and sensors on one pipeline**
✕ A cost centre, start to finish	✓ **Compliance plus commercial insight**

ARCHITECTURE

Built as enterprise software, not a camera add-on

Four layers, deliberately decoupled, so a burst of events or a failed service degrades throughput — never the record.

EDGE	On-site inference on client-owned hardware, taking both camera streams and sensor telemetry.
EVENT BACKBONE	Durable queuing separates detection from processing, absorbing bursts without dropping events on the floor.
DOMAIN SERVICES	Independently scalable services for violation management, audit journaling and human re-identification.
EXPERIENCE	A single gateway serving the operator console, live push notification, analytics and scheduled reporting.

- JWT authentication + RBAC
- Row-level tenant isolation
- Append-only audit journal
- Horizontally scalable services

DEPLOYMENT

Live in weeks, not quarters

A typical single-site rollout. Multi-site programmes run the same sequence per location, in parallel.

Week 1

Site survey

Camera and sensor audit, coverage mapping, and agreement on the SOPs that carry the most operational risk.

Week 2

Install

Edge device commissioned on-site; additional cameras and sensor points mounted and brought online.

Week 3

Configure

Rules mapped source by source, thresholds tuned, recipients and escalation routing set.

Week 4

Validate

A supervised run confirms detection quality and calibrates against real site conditions.

Week 5+

Operate & expand

Live alerting and compliance logging, with further SOPs added by configuration.

Hardware ownership. The edge device is purchased once and titled to you on installation. There is no leasing lock-in, and no dependency on us to keep your own hardware running.

COMMERCIAL MODEL

A transparent investment model

You own the hardware. You pay for what you deploy, and for the rules you need enforced.

Installation

ONE-TIME

Site survey, camera and sensor integration, edge commissioning and initial SOP configuration.

Edge device

ONE-TIME · YOU OWN IT

Purchased once at cost and titled to you on the day it is installed.

Per camera

RECURRING

Platform access, inference capacity, evidence storage and support, per camera online.

Per SOP

RECURRING · SCOPED

Priced by rule complexity — standard detections, telemetry thresholds, or bespoke trained models.

How SOP pricing is set

Cost follows engineering effort, not headcount. A standard presence, PPE or telemetry-threshold check draws on a maintained model and prices at the low end; a bespoke rule requiring data collection, annotation and training sits higher. Every proposal is scoped to your camera count, sensor points and SOP list after the site survey — so the number you are quoted is the number you pay.

Where Alpha Surveillance fits

Video analytics is scaling from $12.7B in 2024 to a projected $37.8B by 2030 — a 19.5% compound annual rate. The same rule engine serves every sector below; only the SOPs change.

Construction & industrial

PPE enforcement, fall-risk zones, equipment proximity and subcontractor accountability.

Warehousing & logistics

Dock safety, forklift–pedestrian conflict, restricted areas and shift headcount.

Retail

Shrinkage, footfall and dwell analytics. Retailers reported an 18% year-on-year rise in shoplifting incidents.

Corporate & facilities

Access-control verification, after-hours presence, visitor and contractor flow.

Healthcare

Hygiene and PPE protocol compliance in clinical, sterile and controlled environments.

Utilities & process plant

Pressure, air quality, thermal and electrical telemetry alongside site access control.

Sources: Grand View Research, Video Analytics Market Size & Share Report (2024–2030); National Retail Federation, The Impact of Retail Theft & Violence (2025).

NEXT STEPS

Start with one site. Prove it in a fortnight.



01 Site walkthrough

We audit your cameras, sensors and coverage, then agree the two or three SOPs that carry the most risk.



02 Pilot

A small camera and sensor set runs live for two weeks. You see real detections on your own site — not a demo reel.



03 Scale

Expand source by source and site by site, on evidence rather than assumption.

Propellus

sabeerah@propellus.co

Alpha Surveillance — Detection, Alerting & Compliance

REFERENCES

Sources

01	**National Safety Council, Injury Facts — Work Injury Costs**	2024 data. injuryfacts.nsc.org/work/costs/work-injury-costs
02	**U.S. Bureau of Labor Statistics — Census of Fatal Occupational Injuries**	2024 summary. bls.gov/news.release/cfoi.nr0.htm
03	**PLOS ONE — "Pirate Stealth or Inattentional Blindness?"**	Security monitoring, experienced and naïve operators; n = 171.
04	**OSHA — Top 10 Most Frequently Cited Standards**	FY 2025 (1 Oct 2024 – 30 Sep 2025). osha.gov/top10citedstandards
05	**OSHA — Penalties**	Maximums effective for citations issued after 15 Jan 2026. osha.gov/penalties
06	**U.S. EPA — PM2.5 National Ambient Air Quality Standards**	24-hour standard 35 µg/m^3, retained in the 2024 final rule.
07	**National Retail Federation — The Impact of Retail Theft & Violence**	2025 edition; survey conducted June–August 2025.
08	**Grand View Research — Video Analytics Market Size & Share Report**	2024 base year, 2025–2030 forecast period.
09	**National Safety Council / NCCI — Workers' Compensation Costs**	Claims for accidents occurring 2022–2023, valued October 2025.

A note on illustrative data. Console views, trend lines, category breakdowns and telemetry threshold values in this deck are illustrative, shown to demonstrate format. They are not measured client results.